PROJECT REPORT

TITLE:

Campus Network Design & Implementation On Cisco Packet

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Abstract:

This project uses Cisco Packet Tracer to simulate a campus-wide network. A hierarchical enterprise architecture is reflected in the design, which divides different departments and buildings into Virtual Local Area Networks (VLANs). VLAN-based segmentation, inter-VLAN routing (router-on-a-stick), dynamic IP allocation via DHCP, wireless connectivity (WLAN) via access points, secure management (SSH/port security), and service hosting (e.g., web, email, DNS) are among the fundamental features. The network offers flexibility, segmentation, and safe resource access for staff, instructors, and students using both wired and wireless devices.

Overview and Goals:

1.1 Background:

To support teaching, administration, research, and student services, modern educational institutions such as universities and colleges need strong network infrastructures. Reliable connectivity, security, scalability, and effective resource sharing are all provided by a well-designed campus network. Without requiring physical hardware, students and network designers can experiment with design, configuration, and management by simulating such a network in Packet Tracer.

1.2 Project Goals:

Create a scalable, hierarchical campus network topology that takes into account the various departments and buildings. Utilize VLANs to logically divide various departments and faculties, allowing for better traffic control and security. To support PCs, laptops, and mobile devices, offer both wired and wireless connectivity (WLAN for mobility). To assist academic and administrative tasks, host necessary services (such as a web server, email server, and DNS). Set up network security features and protocols, such as port security, inter-VLAN routing, DHCP for dynamic IP assignment, and SSH for safe administration. Show how departments and buildings are interconnected while maintaining logical separation.

Architecture & Design of Networks:

2.1 Overview of Topology and Hierarchical Design:

To provide structure, scalability, and management, the network architecture employs a hierarchical (three-tier) paradigm with core, distribution, and access layers.

Core Layer: All major distribution blocks are connected by a central router or layer-3 switch, which also provides routing between VLANs and connection to external servers, links, and the Internet.

Distribution Layer: VLANs, DHCP, inter-VLAN routing, switches and routers for every department or building.

Access Layer: End-device connections: PCs, laptops, wireless access points, and

switches (or wireless APs) that provide endpoint connectivity.

2.2 VLAN and Department Segmentation:

To isolate traffic, enforce security, and streamline administration, each faculty or department (such as Business, Engineering/Computing, Arts, Student Labs, Admin, etc.) is given its own VLAN and matching subnet.

2.3: Service & Server Placement:

On specialized servers linked to the network backbone, the core network provides necessary services (such as web servers, DNS, and email). This preserves centralized management while enabling all VLANs to access services through routing. Switches (or distribution switches) connect wireless access points to the network, allowing WLAN coverage throughout buildings, labs, hostel zones, etc.

Cisco Packet Tracer implementation:

3.1 Tools & Simulation Environment:

Cisco Packet Tracer is a virtual simulation environment with a drag-and-drop interface and CLI configuration support for network devices, including routers, switches, wireless APs, and PCs. Routers, layer-2/3 switches, wireless access points, servers (Web, DNS, and email), PCs, laptops, and cell phones are among the devices utilized.

3.2 Setting Up:

In summary, each department is allotted a VLAN, and switch ports are set up appropriately.
IP addressing and subnetting: each VLAN is given a unique subnet, and the IP scheme is designed to support both present and future hosts while permitting expansion. When communication between VLANs is required, inter-VLAN routing is performed using a router-on-a-stick or layer-3 switch. DHCP configuration: dynamic address distribution for devices to facilitate device management, particularly for wireless devices and student PCs/laptops.
Installing and configuring wireless access points to enable mobile/wireless connectivity for laptops and smartphones around campus is known as wireless LAN (WLAN) configuration.
Security and management: switch port security, secure management access via SSH (if feasible).

3.3 Testing & Verification:

After configuration, simulation was used to verify connectivity: PCs and devices in same VLAN interact correctly. For shared services (such students accessing web/DNS servers), inter-VLAN routing permits allowed access across VLANs. Wireless devices can access services and the Internet via the core router after connecting via APs and obtaining IP addresses via DHCP. For servers and core devices, DHCP and static addressing are operating as intended. Maintaining traffic isolation: When cross-departmental traffic is prohibited, VLAN-based separation stops it.

Findings and Discussion:

Segmentation, scalability, wireless/wired support, and centralized services are all met by the simulated campus network. Organized structure and flexibility for expansion (adding faculty, labs, and additional VLANs) are provided by using hierarchical design and VLANs. Particularly for student devices, mobile devices, and labs, DHCP and WLAN greatly ease user management. The architecture supports both wired and wireless endpoints, matching real-world campus requirements (offices, labs, student mobility). Centralized servers facilitate resource sharing and administration by offering shared services (web, email, DNS).

Limitations: Due to the simulation's nature, real-world difficulties (such as hardware malfunctions, cabling issues, wireless interference, and actual throughput/latency under stress) cannot be accurately duplicated.

Intrusion detection systems, firewalls, monitoring, QoS, and redundancy protocols are examples of advanced security methods that are not used here. Comparable campus-network simulation experiments have similar restrictions.

Simulation limitations limit scalability: re-evaluation may be necessary for growth to very large campuses or many more VLANs/users.

Conclusion:

This project shows how a functional, hierarchical campus network that supports wired and wireless users, offers necessary services, and maintains traffic segmentation via VLANs and appropriate routing can be designed and implemented using Cisco Packet Tracer. The network satisfies the fundamental needs of a contemporary university setting, including administrative manageability, flexibility, service centralization, and user mobility.

The following could be taken into consideration for upcoming improvements:

Use intrusion detection and prevention, network monitoring, and advanced security (firewalls, access control lists, VPNs for remote access).

Increase redundancy through failover methods, numerous routers and switches, and link redundancy. Prioritize traffic using Quality of Service (QoS) (e.g. VoIP, video streaming, academic applications). integration of other services, such as file servers, printer servers, database servers, and VPN/remote access.

Real-world deployment: To assess actual network behavior, performance, and security under load, carry out a physical deployment following simulation.

References:

Preetham N. Varne et al.'s "Campus Network Module using Cisco Packet Tracer Simulator,"

An IJARCCE 2020 article, describes campus network design and implementation using Cisco Packet Tracer.

Journal with peer review Additional publicly accessible GitHub repositories and Packet Tracer projects for university networks.